Q.24 Write a short note on stack and stack pointer.

Q.25 How many types of interrupts in 8051 microcontroller?

Q.26 Write a short note on loop and jump instruction.

Q.27 Explain all the embedded hardware units.

Q.28 What are the examples of embedded system?

Q.29 Write a brief note on serial data input / output in microcontroller.

Q.30 Write a short note on program status word.

Q.31 Write all the addressing modes of 8051 microcontroller.

Q.32 Write a short note on counters and timers in microcontroller.

Q.33 Give a brief note on the present trends in embedded system?

Q.34 Write a short note on advance microcontroller.

Q.35 Write a short note on microcontroller for embedded system.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 What are the arithmetic instructions of 8051? Explain push and pop instruction and call instruction.

Q.37 Write a short note on

a) Pin diagram of microcontroller

b) Architecture of 8051

Q.38 Explain about the advance microcontroller and its applications in detail.

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Roll No. ....

4th Sem / IC
Sub : Microcontroller & Embedded System

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of the following is/are the applications of embedded system?

a) Consumer electronics

b) Automotive system

c) Industrial automation

d) All of the above

Q.2 An 8051 microcontroller is 8 bit micro controller.

a) True

b) False

c) May be true for some applications

d) May be false for some applications

Q.3 Choose the incorrect interrupt signal in 8051.

a) RESET b) Serial interrupt

c) Timer 0 interruput d) Timer 5 interrupt

Q.4 Register A is also known as ______.

a) Accumulator b) B register

c) PSW d) All of the above

Q.5 Which of the following is the immediate instruction?

a) ADD b) MOV

c) None of these d) LXI

Q.6 How many 16 bit counters does 8051 have?

a) 1 b) 2

c) 3 d) None of the above

Q.7 RTO stands for

a) Right time objective

b) Real time operation

c) Recovery time objective

d) Right time operation

Q.8 What is the primary purpose of an embedded system?

a) General purpose computing

b) Performing a single specific task

c) Running heavy simulation software

d) All of the above

Q.9 Which of the following is the best application of 8051 microcontroller?

a) Light sensing & controlling devices

b) Temperature sensing and controlling devices

c) Process control devices

d) All of the above

Q.10 Advanced microcontrollers often feature “sleep” of “Deep sleep” modes to support which type of application?

a) High-speed gaming

b) Replacing the car battery

c) Batter-operated wearable devices

d) Data center servers

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Write any one application of advanced microcontroller.

Q.12 Define microcontroller?

Q.13 The oscillation frequency of 8051 microcontroller is _____ MHz.

Q.14 What is program counter in microcontroller?

Q.15 _____ register is 8 bit register generally used for primary register in arithmetic, logic etc operations.

Q.16 A call instructions transfer program control to a subroutine, pushing the return address onto the stack, and use RET to return of the main program. (True / False)

Q.17 What is an embedded system.

Q.18 Write one application of microcontroller.

Q.19 Expand PSW.

Q.20 Absolute instruction is one of the addressing modes of 8051. (True / False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 List any five applications of microcontroller.

Q.22 What are the advantages of microcontroller over microprocessor?

Q.23 Draw and explain the block diagram of 8051 microcontroller.